FloodSpaceAI — Executive Disaster Situation Report

Incident: Trishuli River Basin Flash Flood / GLOF
Target Flood Date: August 28, 2026
Analysis Area (AOI): Trishuli River Corridor, Nepal `[84.90°E, 27.75°N to 85.25°E, 28.05°N]`
Operational Status: `PARTIAL_RUN_OPTICAL_UNCONFIRMED` (All-Weather SAR Verified; Optical Cloud-Obscured)
Historical Infrastructure Baseline: OpenStreetMap Snapshot `2026-07-27T00:00:00Z` (Strictly Pre-Event)

1. OPERATIONAL OVERVIEW & KEY METRICS

FloodSpaceAI executed end-to-end multi-sensor satellite change detection and road network connectivity modeling to evaluate potential inundation and accessibility disruption across the Trishuli valley.

Analytical Metric	Modeled Value	Analytical Unit	Source Artifact	Scientific Interpretation Notice
Candidate Inundation Extent	**1,286**	pixels	`rasters/final_flood_candidate_v3.tif`	SAR backscatter drop < -1.5 dB; 16.8m x 16.2m resampled grid.
Candidate Surface Water Area	**35.03**	hectares	`rasters/final_flood_candidate_v3.tif`	2D water pooling extent (24.16 ha in EMSR927 overlap); depth not modeled.
Discrete Water Clusters	**27 / 84**	polygons	`vectors/flood_polygons_v3.geojson`	Contiguous standing water bodies (>=2 connected pixels).
Potentially Affected Buildings	**195**	structures	`assets/buildings_affected_v3.geojson`	**Potential exposure only; unconfirmed structural damage.**
Potentially Affected Roads	**115**	features	`assets/roads_affected_v3.geojson`	Centerlines intersecting candidate water polygons.
Exposed Highway Length	**59.26**	kilometers	`assets/roads_affected_v3.geojson`	Cumulative exposed highway length; not destroyed pavement.
Potentially Affected Bridges	**13**	bridges	`assets/bridges_affected_v3.geojson`	2D overlay; vertical bridge deck clearance not modeled.
Disrupted Network Edges	**3,748**	segments	`connectivity/settlement_connectivity_v4.json`	Routable road edges severed in topological routing graph.
Potentially Cut-Off Settlements	**18**	settlements	`connectivity/settlements_cutoff_v4.geojson`	**Modelled road network disconnection; not physical isolation.**

2. PRIORITY SETTLEMENTS WITH MODELED VEHICULAR ACCESS LOSS

Topological Dijkstra shortest-path pathfinding from 589 settlements to municipal town/city hubs (Bidur, Galchhi, Belkotgadhi) identified **18 settlements** that lost all vehicular access routes:

Settlement Name	OSM Classification	Nearest Municipal Hub	Pre-Flood Status	Modeled Post-Flood Access
Betrawati (बेत्रावती)	Hamlet	Bidur (बिदुर)	Connected	**Disconnected (Bridge/Highway Cut)**
Bhaise (भैंसे)	Hamlet	Bidur (बिदुर)	Connected	**Disconnected (Riverbank Overwash)**
Simle (सिम्ले)	Hamlet	Bidur (बिदुर)	Connected	**Disconnected (Highway Segment Severed)**
Devighat (देवीघाट)	Hamlet	Belkotgadhi (बेलकोटगढी)	Connected	**Disconnected (Bridge Approach Cut)**
Dhaibung (धइबुङ)	Hamlet	Bidur (बिदुर)	Connected	**Disconnected (Corridor Inundation)**
Salim Bhitta (सालिम भित्ता)	Hamlet	Bidur (बिदुर)	Connected	**Disconnected (Valley Floor Submergence)**
Dhungye (ढुङ्गे)	Hamlet	Bidur (बिदुर)	Connected	**Disconnected (Access Road Blocked)**
Uttargaya (उत्तरगया)	Village	Bidur (बिदुर)	Connected	**Disconnected (Primary Feeder Severed)**
Kalika (कालिका)	Village	Bidur (बिदुर)	Connected	**Disconnected (Route Impassable)**

> **Responder Guidance:** Modelled cut-offs reflect vehicular highway disruption. Emergency teams should prioritize foot patrols, unmapped pack trails, boat crossings, or helicopter reconnaissance for these locations.

3. MANDATORY SCIENTIFIC LIMITATIONS & ATTRIBUTIONS

Exposure vs. Confirmed Damage: All figures represent modeled potential inundation and spatial intersection with OpenStreetMap infrastructure footprints. They **do not confirm physical destruction, structural collapse, or water ingress**. In-situ engineering inspection is required before declaring structural losses.
Topological Disruption vs. Physical Isolation: Cut-off settlements represent vehicular graph disconnects. Footpaths, mule trails, river ferries, or aerial access routes may remain open.
Sensor Limitations & Cloud Cover: Post-event Sentinel-2 optical imagery had 75.9% mean cloud obscuration. Inundation is derived primarily from all-weather Sentinel-1 C-band synthetic aperture radar and remains optical-unconfirmed.
Independent Validation Isolation: Copernicus Emergency Management Service EMSR927 debris-flow footprints (1,068.36 ha, 1.01% IoU geometric overlap) are maintained strictly as an independent post-hoc co-location check isolated from pipeline inputs; this comparison is limited and cannot establish flood-detection accuracy by itself.
Attribution: Contains modified Copernicus Sentinel data 2026. Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved. © OpenStreetMap contributors.